TerraTransfer

APPLICATION PACKAGE · TYP

Water Level & Conductivity.

AQUATOS WEB LTX · PIEZO 0312 · TYPE ES

DESCRIPTION

The **Water Level & Conductivity application package** combines the flagship data logger **Aquatos Web LTX** with the high-precision piezo pressure sensor **Type 0312** and the four-electrode conductivity sensor **Type ES** to form a self-sufficient monitoring station for groundwater, surface water and critical infrastructure.

Both sensors run on the logger's single **SDI-12 V1.3** bus — one sensor cable, no additional signal converters. The logger captures **water level, water temperature, electrical conductivity and fluid temperature** synchronously and transmits the data via **LTE-M** (with 2G fallback) to the **Sensormanager** web portal.

End-to-end encryption using **AES-128-CBC** (recommended by the German Federal Office for Information Security) and signed firmware make the package suitable for regulated monitoring networks and critical infrastructure. On-site configuration is done via **Bluetooth 5** — no app, no cable, no SD card.

APPLICATIONS

- Groundwater monitoring with salinity control (coastal regions, bank filtrate)
- Surface water — conductivity profiles in rivers, streams and lakes
- Drinking water protection and early warning of pollutant ingress
- Agriculture and irrigation — salinity of irrigation water
- Landfill and legacy-site monitoring, leachate measurement
- Research monitoring networks with high temporal resolution

DATA LOGGER · TYPE 1500

PRESSURE SENSOR · TYPE 0312

EC + TEMP · TYPE ES

KEY FEATURES

- **Three measurands** — level, EC, temperature
- **SDI-12 bus**: both sensors on a single input
- **Up to 120 dS/m** EC · four-electrode principle
- **±0.1% FS** long-term stability of the piezo
- **LTE-M** + 2G fallback · **AES-128-CBC**
- Up to **10 years** battery life
- **IP68** · operation from −25 °C to +85 °C
- Local configuration via **Bluetooth 5**

Technical Data & Interfaces

AQUATOS WEB LTX · PIEZO 0312 · TYPE ES

AQUATOS WEB LTX · DATA LOGGER

Input channels	24 digital channels · SDI-12 V1.3 · max. 24 SDI-12 sensors
Output signal	LTE-M / 2G (bidirectional) · Bluetooth 5 (BLE) locally
Antenna	4G antenna · FME connector · external
Encryption	AES-128-CBC (BSI-recommended) · BLE challenge-response auth
Power supply	Lithium battery · up to 10 years operation (>3 million readings)
Operating temperature	−25 °C to +85 °C · no icing
Ingress protection	IP68 (24 h at 1 m submersion depth)
Software	Sensormanager web portal · BLX.JS app (Web Bluetooth, Chrome / Edge)
Conformity	CE (RED 2014/53/EU, EMC 2014/30/EU) · RoHS

PIEZO PRESSURE SENSOR · TYPE 0312

Measurement principle	Piezoresistive · oil-filled stainless-steel cell · laser-welded
Measurement ranges	5 · 10 · 20 · 30 · 100 mH_2O (project-specific)
Long-term stability	typ. ±0.1% FS per year
Zero offset (tare)	On-site zero calibration via BLX.JS
Temperature measurement	Integrated · accuracy ±0.1 °C
Housing / diaphragm	AISI 316 Ti · stainless-steel or titanium diaphragm
Interface	SDI-12 V1.3 (VCC · Data · GND) + Bluetooth 5
Power supply	2.8 V – 14 V · sleep current < 10 µA
Ingress protection	IP68 · permanently submersible

Conductivity · Type ES & Ecosystem

TECHNICAL DATA · SOFTWARE · SCOPE OF DELIVERY

TYPE ES · CONDUCTIVITY + TEMPERATURE

EC principle	Four-electrode array · stainless steel · low contamination sensitivity
EC range	0 – 120 dS/m
EC resolution / accuracy	0.001 dS/m · ±0.01 dS/m or ±10% (whichever is greater)
Temperature range	−40 °C to +60 °C · resolution 0.1 °C · ±1 °C
Interface	SDI-12 V1.3 · user-selectable SDI-12 address · TT SDI-12 adapter
Power supply	3.6 – 15 V DC · 20–28 mA (measurement) · 0.03 mA (sleep)
Dimensions	Length 10.9 cm · diameter 2.4 cm · thread ½″ NPT
Cable	Custom length per order · max. 75 m

Sensormanager

Web portal with database, hydrographs, maps, alarms and API. Browser and mobile web app. Data retention up to 10 years, GDPR-compliant.

BLX.JS app

Browser-based configuration via Web Bluetooth at **https://ltxdata.net/ttblx/** — no installation required. Chrome and Edge (desktop / Android).

Scope of delivery

Aquatos Web LTX, Piezo 0312 (range per order), Type ES with cable in custom length, SDI-12 splitter, 4G antenna, lithium battery.

TerraTransfer

MADE IN GERMANY

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SUBJECT TO CHANGE · WEIGHTS AND DIMENSIONS ARE APPROXIMATE

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